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$$f : \Omega \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^2 : (x, y) \rightarrow \left(e^{xy}, \ln \frac{x}{y} \right)$$

$$Df(x, y) = \begin{pmatrix} ye^{xy} & xe^{xy} \\ \frac{1}{x} & \frac{-1}{y} \end{pmatrix}$$

$$\det(Df(x, y)) = -2e^{xy} \neq 0$$

$$Df^{-1}(x, y) = \frac{1}{-2e^{xy}} \cdot \begin{pmatrix} \frac{-1}{y} & -xe^{xy} \\ \frac{-1}{x} & ye^{xy} \end{pmatrix} = \begin{pmatrix} \frac{1}{2ye^{xy}} & \frac{x}{2} \\ \frac{1}{2xe^{xy}} & \frac{-y}{2} \end{pmatrix}$$

References